

Week 1

DiskMath 16:15 ~ 18:00 CHN E42

About me:

Zijun Wan zijwan@student.ethz.ch

3rd semester CS Bachelor

Language:

English

Exercise Session Layout

- Lecture Recap
- Previous Bonus problem
- (Optional) non-bonus exercises
- Extra exercises / exam problems

45 min
15 min
45 min

1h 30
15 min

Logistics:

DiskMath CHN E42

WhatsApp-Gruppe



- asking questions
- polls on topics that were hard for you
↳ discuss more in next session
- notes from exercise sessions (website)

Bonus: Oral exercises

- 2x 15-min sessions

- present the bonus problem of current week
- **NO AI !!! Individual work !!!**
- Allowed to bring notes.
- Understand each step of your solution
- Hints ✓ (May lead to deduction of points)
- max. 6 points for each oral session

If time allows:

Bonus questions — make up for other questions

- max. 0.25 bonus for exam
- Absence? Find substitute!

Exam: 3h , very hard

- 6 A4 pages Cheat Sheet

Tips:

- Read the script
- Do the exercises (including non-bonus)
 - Try to solve on your own. Don't give up too early.
 - Good practice for exam: Write down all your steps
- Clarify anything unclear
 - Internet, LLM, ...
 - Ask prof, TA, classmates, ...
- If everything feels overwhelming,
don't worry! It gets better!

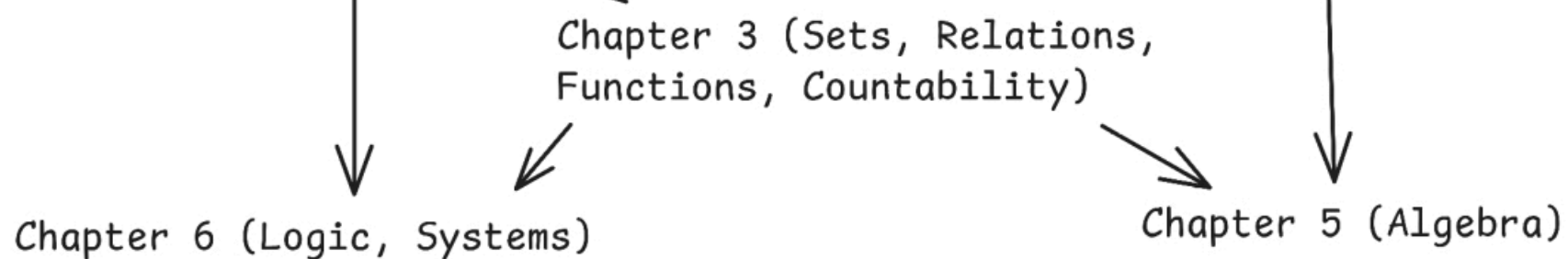
Lecture Overview:

Chapter 1 (Intro)

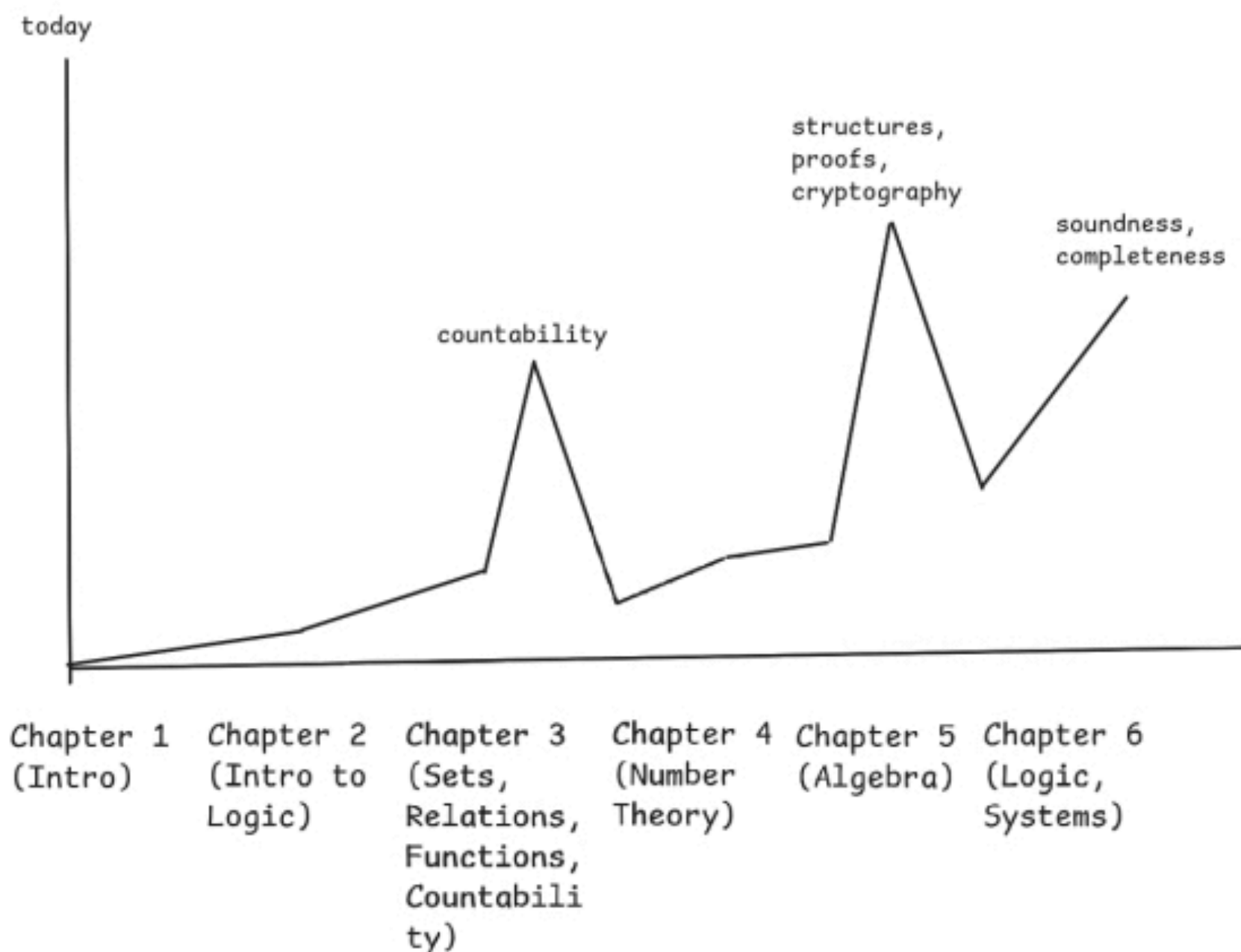


Chapter 2 (Intro to Logic)

Chapter 4 (Number Theory)



Perceived Difficulty:



Lecture Recap

Mathematical Statements , True or False

- "What time is it now?" X
- " $2 \cdot 3 = 6$ " ✓ True
- "6 is a prime number." ✓ False
- " $x > 5$ " X

Definition 2.1. A *mathematical statement* (also called *proposition*) is a statement that is true or false in an absolute, indisputable sense, according to the laws of mathematics.

Implication

" $S \Rightarrow P$ " : Statement
 $\swarrow \quad \searrow$
 Statements

Def: S true, P false : " $S \Rightarrow P$ " false
 otherwise: " $S \Rightarrow P$ " true

S	P	$S \Rightarrow P$
F	F	T
F	T	T
T	F	F
T	T	T

Intuition:

You missed the train $\Rightarrow$ You will be late to the lecture

If left side false, then we don't know abt. right side

You didn't miss the train, then both possible:

- Got lost in HG
- Found lecture hall

If left side true, then right side must be true

You missed the train, then you will be late.

Cases:

Didn't miss train^F and not late. T
 Didn't miss train^F and late. T
 Missed train^T and not late. F
 Missed train and late. T

S	P	$S \Rightarrow P$
F	F	T
F	T	T
T	F	F
T	T	T

Examples:

- $2 \cdot 3 = 6$ ^T $\Rightarrow$ 6 is not prime ^T T
- 6 is prime ^F $\Rightarrow$ $2 \cdot 3 = 6$ ^T

Implication $\neq$ Causality!

$\Rightarrow$: In proofs

$S \Rightarrow T$: T follows from S (with justification)

$2 \cdot 3 = 6 \Rightarrow 6$ is not prime

$\Rightarrow 6$ is not prime for all $a \in \mathbb{N}$

$\Rightarrow \dots$

But: "3 is prime" $\Rightarrow$ " π is irrational"
without justification.

Propositional Logic

Symbols: $A, B, \dots$

Operators: $\wedge$, $\vee$, $\rightarrow$, $\neg$.
 and or if then not

Formula: $A, B, A \vee B, A \rightarrow B, \neg B,$
 $\neg(A \wedge B) \rightarrow C, \dots$

Truth Tables:

Example: $\neg(A \vee B) \rightarrow A$

A	B	$A \vee B$	$\neg(A \vee B)$	$\neg(A \vee B) \rightarrow A$
0	0	0	1	0
0	1	1	0	1
1	0	1	0	1
1	1	1	0	1

$A \rightarrow B$ is the same as $\neg A \vee B$

A	B	$A \rightarrow B$	$\neg A \vee B$
0	0	1	1
0	1	1	1
1	0	0	0
1	1	1	1

$A \rightarrow B \equiv \neg A \vee B$

Comparisons

$$3x = 3 \Rightarrow x = 1$$

Derives one statement
from another statement

Makes a statement
about formulas.

Derives one formula

$\Rightarrow$

$\Rightarrow$

$\rightarrow$

from another formula
 $\models$

Combines two statements
to form a new statement

$$0=1 \Rightarrow 2 \cdot 3=6$$

Combines two formulas
to form a new formula

$$A \rightarrow \neg B$$

$F \models G$: Whenever F is true, G is also true

Example: $A \wedge B \models A$

A	B	$A \wedge B$	$(A \wedge B) \rightarrow A$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	1

$F \not\models G$

$F \rightarrow G$

0
1
1
1
1

Brackets:

$(A \wedge B) \rightarrow A$

necessary

$(A \wedge B) \vee C$

necessary

$(\neg A) \vee B$

not necessary,
convention

$F \models G$: $F \rightarrow G$ is a tautology (statement)

Analog:

$\Leftrightarrow$

$\Leftrightarrow$

$\leftrightarrow$

$\equiv$

"=" vs " $\equiv$ "

$$A \vee B \equiv B \vee A, \quad A \vee B \neq B \vee A$$

Compare	statements	vs	formula
truth value	fixed		variable
ex.	$0=1$		$A \wedge (B \vee C)$

Find mistakes:

$S \wedge T$, S, T statements

S and T

$A \Rightarrow B$, A, B formulas

$A \rightarrow B$

Tautology & Satisfiability

Tautology: $T \quad A \vee (\neg A) \equiv T$

Every entry in the truth table is 1.

Every entry in the truth table is 0.

$$T \Rightarrow \dots \Rightarrow T$$

i) $F_1 = \neg B \rightarrow \neg A$ **ii)** $F_2 = (A \wedge B) \vee (\neg A \wedge \neg B)$

2. (★) Using only the propositions A, B and logical operators, write down formulas corresponding to the following sentences:

i) F_3 : "Bob neither forgot to pay his rent nor is he getting evicted."

ii) F_4 : "Bob either forgot to pay his rent or he is getting evicted, but not both."

3. (★ ★) For both formulas F_3, F_4 , write down their negations both as sentences and formally as formulas.

2 i) $\neg A \wedge \neg B$ $\neg(A \vee B)$
 ii) $(A \vee B) \wedge \underbrace{(\neg(A \wedge B))}_{\text{not both}}$
 $(A \wedge \neg B) \vee (\neg A \wedge B)$

3. $F_3 = A \vee B$

$F_4 = A \leftrightarrow B$

$(\neg A \wedge \neg B) \vee (A \wedge B)$

F_4 : forgot x evicted
 x forgot evicted

$A \wedge B$

$\neg F_4$: forget evicted
 x forget x evicted

Exercise 1.6 Simplifying a Formula (★)

Consider the propositional formula

$$F = (C \vee ((A \vee C) \wedge (A \vee B))) \wedge (B \rightarrow C).$$

Give a formula G that is equivalent to F , but in which each atomic formula A, B , and C appears at most once. Prove that $F \equiv G$ by providing a sequence of equivalence transformations with at most 6 steps.

Expectation. Your proof should be in the form of a sequence of steps, where each step consists of applying the definition of $\rightarrow$ (that is $F \rightarrow G \equiv \neg F \vee G$), one of the rules given in Lemma 2.1 of the lecture notes, or one of the following rules: $F \wedge \neg F \equiv \perp$, $F \wedge \perp \equiv \perp$, $F \vee \perp \equiv F$, $F \vee \neg F \equiv \top$, $F \wedge \top \equiv F$, and $F \vee \top \equiv \top$. Lemma 2.1 states rules involving propositional symbols, but you may apply those rules at the level of formulas (see Section 2.3.5 of the lecture notes). For this exercise, associativity is to be applied as in Lemma 2.1 3). Each step of your proof should apply a **single** rule **once** and state **which** rule was applied.

Lemma 2.1.

1) $A \wedge A \equiv A$ and $A \vee A \equiv A$ (idempotence);

2) $A \wedge \neg A \equiv \perp$, $\neg A \wedge A \equiv \perp$, $A \vee \neg A \equiv \top$, $\neg A \vee A \equiv \top$ (complementarity);

- 2) $A \wedge B \equiv B \wedge A$ and $A \vee B \equiv B \vee A$ (commutativity of $\wedge$ and $\vee$);
- 3) $(A \wedge B) \wedge C \equiv A \wedge (B \wedge C)$ and $(A \vee B) \vee C \equiv A \vee (B \vee C)$ (associativity);
- 4) $A \wedge (A \vee B) \equiv A$ and $A \vee (A \wedge B) \equiv A$ (absorption);
- 5) $A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$ (first distributive law);
- 6) $A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$ (second distributive law);
- 7) $\neg\neg A \equiv A$ (double negation);
- 8) $\neg(A \wedge B) \equiv \neg A \vee \neg B$ and $\neg(A \vee B) \equiv \neg A \wedge \neg B$ (de Morgan's rules).

$$F = C \vee (A \vee (B \wedge C)) \wedge (B \rightarrow C) \\ \wedge (\neg B \vee C)$$

$$(C \vee ((A \vee C) \wedge (A \vee B))) \wedge (B \rightarrow C)$$

$$\equiv (C \vee (A \vee (C \wedge B))) \wedge (B \rightarrow C)$$

$$\equiv (C \vee ((C \wedge B) \vee A)) \wedge (B \rightarrow C)$$

$$\equiv ((C \vee (C \wedge B)) \vee A) \wedge (B \rightarrow C)$$

$$\equiv (C \vee A) \wedge (B \rightarrow C)$$

$$\equiv (C \vee A) \wedge (\neg B \vee C)$$

$$\equiv (C \vee A) \wedge (C \vee \neg B)$$

$$\equiv C \vee (A \wedge \neg B)$$

distr.

comm.

abs.

absorption

$\neg \rightarrow$

comm

distr.

1.5 Function Tables of Special Symbols

(8 Points)

We define two binary logical operators $\heartsuit$ and $\square$ as follows:

A	B	$A \heartsuit B$
0	0	1
0	1	0
1	0	0
1	1	0

A	B	$A \square B$
0	0	0
0	1	0
1	0	1
1	1	0

a) (*) Compute the function table for the following formula:

$$F = (A \square B) \heartsuit (B \square A).$$

b) (**) Find a formula G involving only $\heartsuit$ and $\square$ that has the following function table:

A	B	G
0	0	1
0	1	1
1	0	0
1	1	1

No justification is required.

b)	$A \heartsuit B$	$B \heartsuit A$	$A \square B$	$B \square A$
	1	1	0	0
	0	0	0	1
	0	0	1	0
	0	0	0	0

$$G = (A \square B) \heartsuit (A \square B)$$

Exercise 1.4 Logical Equivalence via Function Tables

1. (*) Compute the function table for the following formula:

$$(B \rightarrow C) \rightarrow (\neg(A \rightarrow C) \wedge \neg(A \vee B)).$$

2. (**) Give another formula that is equivalent to the formula from Subtask a), but in which each of the propositional symbols appears at most once.